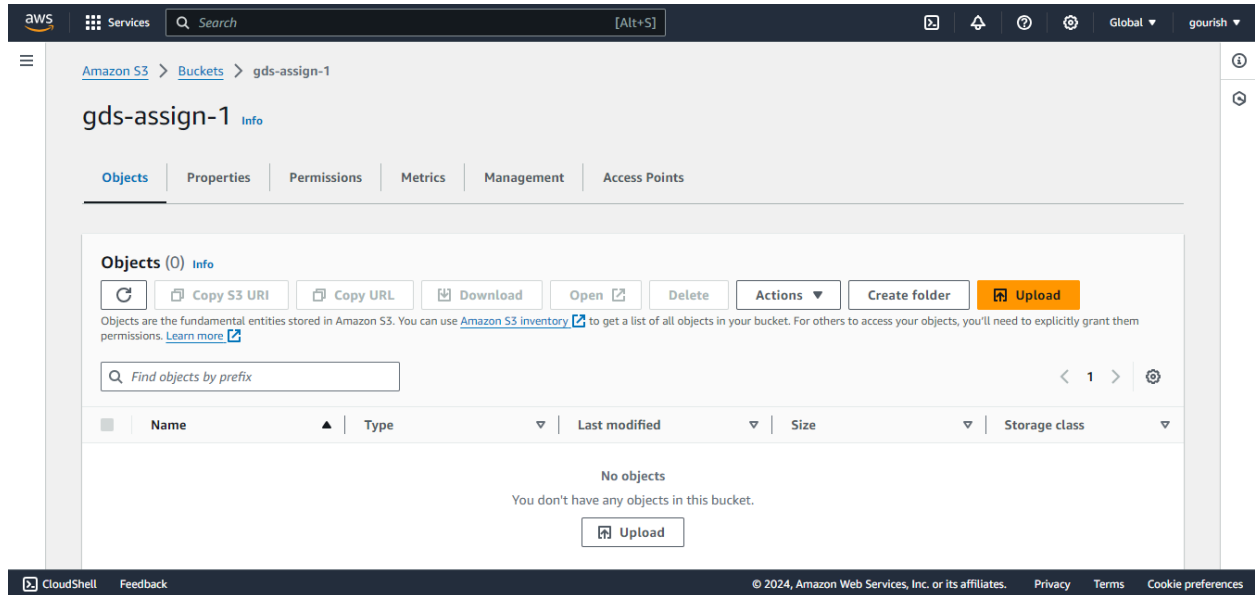


Assignment 1: AWS lambda for Monitoring s3 Uploads

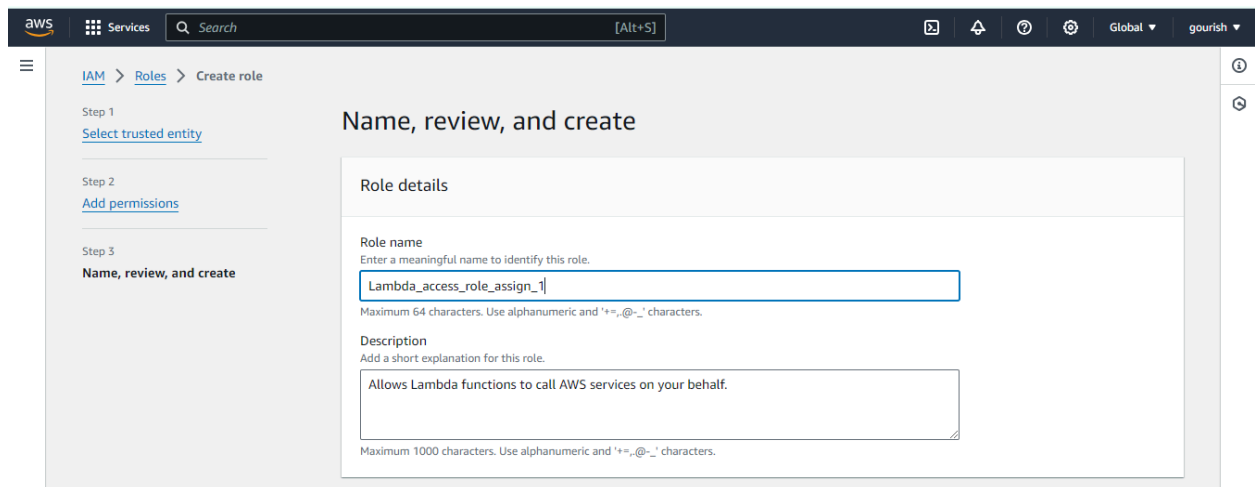
S3 Bucket Setup

- Created a bucket “gds-assign-1”



IAM Role Creation

- Created a IAM Role for AWS Lambda.
- Added S3 full access permission and cloudwatch full access



Maximum 1000 characters. Use alphanumeric and '+', '@', '-' characters.

Step 1: Select trusted entities Edit

Trust policy

```

1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Action": [
7-         "sts:AssumeRole"
8-       ],
9-       "Principal": {
10-        "Service": [
11-          "lambda.amazonaws.com"
12-        ]
13-      }
14-    ]
15-  }
16- }

```

Step 2: Add permissions Edit

Permissions policy summary

Policy name	Type	Attached as
AmazonS3FullAccess	AWS managed	Permissions policy
CloudWatchFullAccess	AWS managed	Permissions policy

Lambda Function Development

- Created Lambda function "gds-lambda-assign-1"
- Added the IAM role which was created in the previous step.

Function name
Enter a name that describes the purpose of your function.
gds-lambda-assign-1
Use only letters, numbers, hyphens, or underscores with no spaces.

Runtime Info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
Python 3.8 C

Architecture Info
Choose the instruction set architecture you want for your function code.
☒ x86_64
☐ arm64

Permissions Info
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role
Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☐ Create a new role with basic Lambda permissions
☒ Use an existing role
☐ Create a new role from AWS policy templates

Existing role
Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.
Lambda_access_role_assign_1 C
View the [Lambda_access_role_assign_1](#) role on the IAM console.

c.

[Lambda](#) > [Functions](#) > gds-lambda-assign-1

gds-lambda-assign-1 Throttle Copy ARN Actions

▼ Function overview Info Export to Application Composer Download

Diagram Template

 **gds-lambda-assign-1**

 **Layers** (0)

Description
-

Last modified
2 hours ago

Function ARN

d.

- e. Added code as per requirement for the assignment.



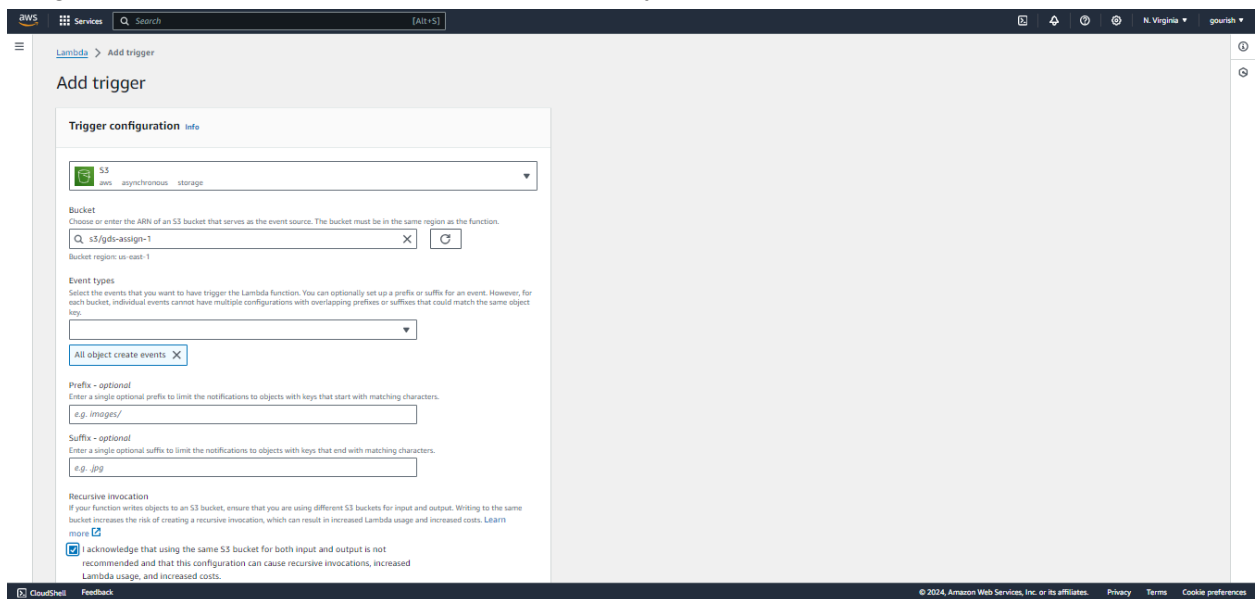
The screenshot shows a code editor with a file named `lambda_function.py`. The code is a Python lambda function handler that processes S3 events. It imports `json` and defines `lambda_handler(event, context)`. The function prints the event and context, then checks if the event is a record. If it is, it prints the bucket name, file name, and file size. If the file size is greater than 10MB, it returns a 413 status code with a message 'File size exceeds 100MB'. Otherwise, it returns a 200 status code with the bucket name, file name, and file size.

```
1 import json
2
3 def lambda_handler(event, context):
4     print('event', event)
5     print('context', context)
6     print("Records", event['Records'][0])
7     print("S3 bucket name", event['Records'][0]['s3']['bucket']['name'])
8     print("S3 file name", event['Records'][0]['s3']['object']['key'])
9     print("S3 file size", event['Records'][0]['s3']['object']['size'])
10    if (event['Records'][0]['s3']['object']['size'] >= 10*8):
11        return {
12            'statusCode': 200,
13            'body': {
14                'Bucket Name': event['Records'][0]['s3']['bucket']['name'],
15                'S3 File Name': event['Records'][0]['s3']['object']['key'],
16                'S3 file size': event['Records'][0]['s3']['object']['size']
17            }
18        }
19    else:
20        return {
21            'statusCode': 413,
22            'body': 'File size exceeds 100MB'
23        }
24
25
```

- f.

S3 to Lambda Integration.

- a. Integrated S3 with lambda to create events activity.



The screenshot shows the AWS Lambda console 'Add trigger' page for an S3 bucket. The trigger is configured to trigger on 'All object create events'. The bucket is `s3/gds-assign-1` in the `us-east-1` region. The event types are set to 'All object create events'. The prefix is optional and set to `images/`. The suffix is optional and set to `.jpg`. The recursive invocation checkbox is checked, indicating that the function can write to the same bucket as it reads from.

Trigger configuration

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.
Bucket: `s3/gds-assign-1`
Bucket region: `us-east-1`

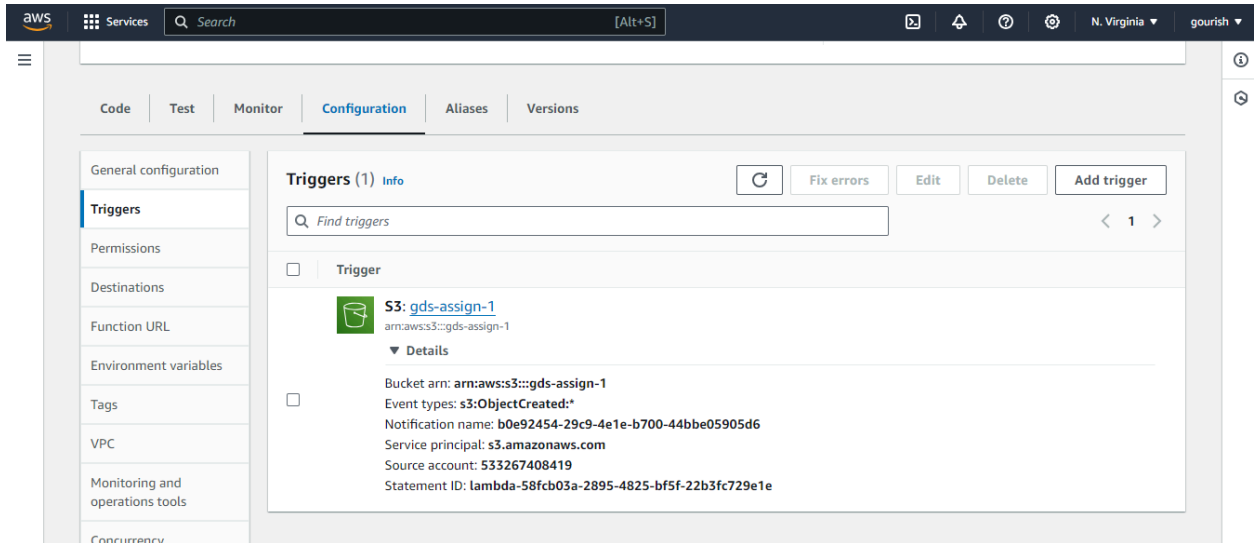
Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.
Event types: `All object create events`

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters.
Prefix: `images/`

Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters.
Suffix: `.jpg`

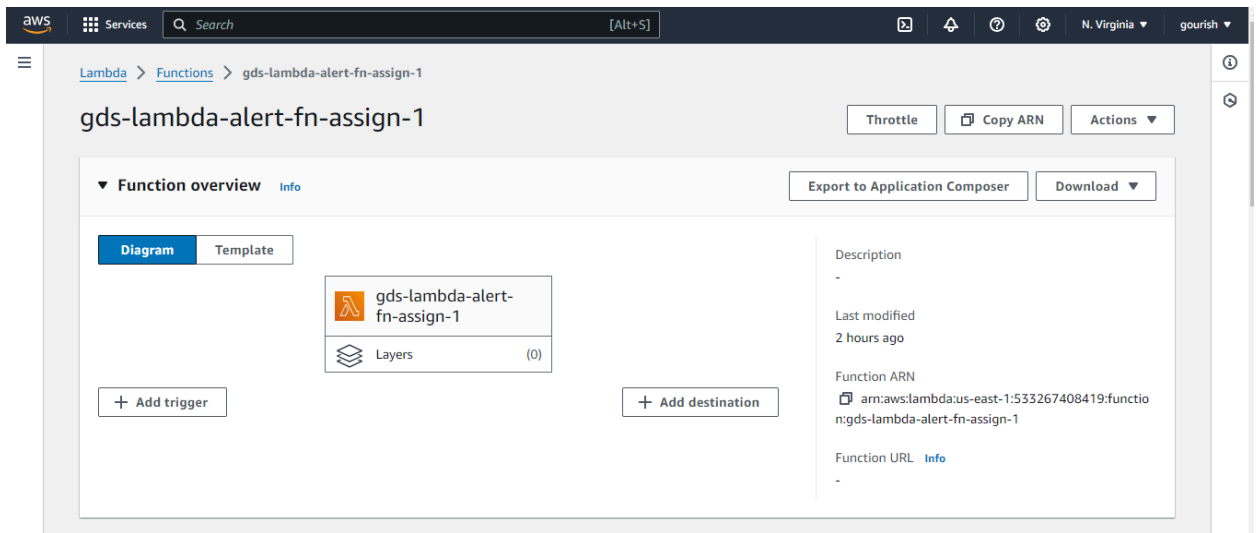
Recursive invocation
If your function writes objects to an S3 bucket, ensure that you are using different S3 buckets for input and output. Writing to the same bucket increases the risk of creating a recursive invocation, which can result in increased Lambda usage and increased costs. [Learn more](#)
☒ I acknowledge that using the same S3 bucket for both input and output is not recommended and that this configuration can cause recursive invocations, increased Lambda usage, and increased costs.

- b.



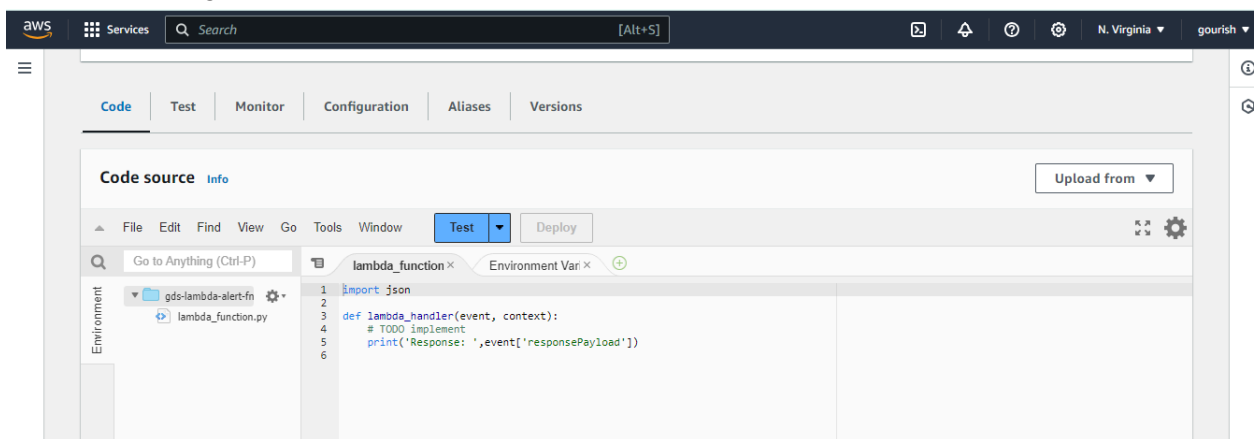
c.

d. Created another lambda function as destination to view the response.



e.

f. Code for printing the response.



g.

Validation

Case 1:

a. Uploaded file less than 100MB.

aws

Services

Search

[Alt+S]

Global

gourish

Amazon S3

Buckets

gds-assign-1

anime.csv

anime.csv

Info

Copy S3 URI

Download

Open

Object actions

Properties

Permissions

Versions

Object overview

Owner
gourish.ganta777

AWS Region
US East (N. Virginia) us-east-1

Last modified
March 7, 2024, 13:18:01 (UTC+05:30)

Size
914.5 KB

Tvne

S3 URI
s3://gds-assign-1/anime.csv

Amazon Resource Name (ARN)
arn:aws:s3::gds-assign-1/anime.csv

Entity tag (Etag)
f460766db19c648cb8c6886f905ba8f6

Object URL
https://gds-assign-1.s3.amazonaws.com/anime.csv

b.

c. Response from 'gds-lambda-assign-1'

2024-03-07T13:18:02.050+05:30

Records

{'eventVersion': '2.1', 'eventSource': 'aws:s3', 'awsRegion': 'us-east-1', 'eventTime': '2024-03-07T07:48:00.318Z', 'eventName': 'ObjectCreated:Put', 'userIdentity': {'principalId': 'ACTOSK6T43A0N'}, 'requestParameters': {'sourceIPAddress': '157.48.192.149'}, 'responseElements': {'x-amz-request-id': 'FRM8724N33VMAXAT', 'x-amz-id-2': 'o5SpLV0i0CYVv650m7gxhhXoLlqIFJmPgamt15AMDbQonXHimQ2gfhHKOititJEN0q2Y3UfAsU3VEUvLhpCRDPqT8YubHx'}, 's3': {'s3SchemaVersion': '1.0', 'configurationId': 'b0e92454-29c9-4e1e-b700-44bbe05905d6', 'bucket': {'name': 'gds-assign-1', 'ownerIdentity': {'principalId': 'ACTOSK6T43A0N'}, 'arn': 'arn:aws:s3::gds-assign-1'}, 'object': {'key': 'anime.csv', 'size': 936463, 'eTag': 'f460766db19c648cb8c6886f905ba8f6', 'sequencer': '0065E9713037E086B4'}}}

Copy

2024-03-07T13:18:02.050+05:30

S3 bucket name

gds-assign-1

Copy

2024-03-07T13:18:02.050+05:30

S3 file name

anime.csv

Copy

2024-03-07T13:18:02.050+05:30

S3 file size

936463

Copy

2024-03-07T13:18:02.050+05:30

END

RequestId: 1cb38360-284a-4571-9363-81c361a0f1a5

Back to top

d.

e. Response from 'gds-lambda-alert-fn-assign-1'

2024-03-07T13:16:19.003+05:30

Response: {'statusCode': 200, 'body': {'Bucket Name': 'gds-assign-1', 'S3 File Name': 'anime.csv', 'S3 file size': 936463..

Copy

2024-03-07T13:16:19.004+05:30

END

RequestId: 30c80ccd-ea9f-4490-a111-0f1245ac8494

f.

Case 2:

- a. Upload file greater than 100MB.

The screenshot shows the AWS S3 console interface for the bucket 'gds-assign-1'. The file 'job_skills.csv' is selected. The 'Properties' tab is active, displaying the following information:

- Object overview**
- Owner:** gourish.ganta777
- AWS Region:** US East (N. Virginia) us-east-1
- Last modified:** March 7, 2024, 13:21:35 (UTC+05:30)
- Size:** 641.6 MB (highlighted in yellow)
- TVpe:** (partially visible)
- S3 URI:** s3://gds-assign-1/job_skills.csv
- Amazon Resource Name (ARN):** arn:aws:s3::gds-assign-1/job_skills.csv
- Entity tag (Etag):** f61cabb333c456b7b7873afdec889957-40
- Object URL:** https://gds-assign-1.s3.amazonaws.com/job_skills.csv

- b.
- c. Response from 'gds-lambda-assign-1'

The screenshot shows the AWS Lambda console interface for the function 'gds-lambda-assign-1'. The response is displayed in a log format with the following entries:

- 2024-03-07T13:23:29.675+05:30** context <__main__.LambdaContext object at 0x7fc8ee47cfd0>
- 2024-03-07T13:23:29.675+05:30** Records {'eventVersion': '2.1', 'eventSource': 'aws:s3', 'awsRegion': 'us-east-1', 'eventTime': '2024-03-07T07:53:28.893Z', 'eventName': 'ObjectCreated:CompleteMultipartUpload', 'userIdentity': {'principalId': 'ACTOSK6T43A0N'}, 'requestParameters': {'sourceIPAddress': '49.205.249.123'}, 'responseElements': {'x-amz-request-id': 'YGT8EFYHBZJ3GJ4N', 'x-amz-id-2': 'j/8JT+XvOyDInUS/xK1D5tpT7UTzIGHG7yTg/2+WTG05g2t8DSY5ybrv3obk0Kvg1jZDhhgIVXghCluEme6u8BQAXvj5GP'}, 's3': {'s3SchemaVersion': '1.0', 'configurationId': 'b0e92454-29c9-4e1e-b700-44bbe05905d6', 'bucket': {'name': 'gds-assign-1', 'ownerIdentity': {'principalId': 'ACTOSK6T43A0N'}, 'arn': 'arn:aws:s3:::gds-assign-1'}, 'object': {'key': 'job_skills.csv', 'size': 672718092, 'eTag': 'f61cabb333c456b7b7873afdec889957-40', 'sequencer': '0065E9720631C1829D'}}}
- 2024-03-07T13:23:29.675+05:30** S3 bucket name gds-assign-1
- 2024-03-07T13:23:29.675+05:30** S3 file name job_skills.csv
- 2024-03-07T13:23:29.675+05:30** S3 file size 672718092
- 2024-03-07T13:23:29.684+05:30** END RequestId: b1cc62d5-c4da-48f8-860c-c379f899e7b1

- d.
- e. Response from 'gds-lambda-alert-fn-assign-1'

The screenshot shows the AWS Lambda console interface for the function 'gds-lambda-alert-fn-assign-1'. The response is displayed in a log format with the following entries:

- 2024-03-07T13:23:30.143+05:30** START RequestId: c1e2b21b-6561-4c39-a48c-607739606cdd Version: \$LATEST
- 2024-03-07T13:23:30.144+05:30** Response: {'statusCode': 413, 'body': 'File size exceeds 100MB'}
- 2024-03-07T13:23:30.145+05:30** END RequestId: c1e2b21b-6561-4c39-a48c-607739606cdd

- f.